MODULE 5: PARTIAL DIFFERENTIAL EQUATIONS (CO: 5)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with VIT BAMAT101 Official Syllabus, VIT Department of Mathematics Question Bank, and VIT FAT Past Papers

Scope of This Textbook (All 8 Syllabus Topics): This textbook covers every topic in Module 5 of your syllabus screenshot: (1) Formation of partial differential equations, (2) Solutions of standard types of first-order partial differential equations–1, (3) Solutions of standard types of first-order partial differential equations–2, (4) Lagrange's linear equation and problems–1, (5) Lagrange's linear equation and problems–2, (6) Solution of a linear first-order PDE by separation of variables, (7) Lagrange's equation for flow modelling, and (8) Separation of variables for analyzing CPU temperature.

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1 VIT FAT Exam Blueprint & Master Notation Sheet for Module 5

Throughout Module 5, $z = z(x, y)$ is a function of two independent variables (x, y), and standard Monge notation is used universally in VIT exams:

$$\boxed{p = \frac{\partial z}{\partial x}, \qquad q = \frac{\partial z}{\partial y}, \qquad r = \frac{\partial^2 z}{\partial x^2}, \qquad s = \frac{\partial^2 z}{\partial x \partial y}, \qquad t = \frac{\partial^2 z}{\partial y^2}.}$$

In VIT Final Assessment Tests (BMAT102L / BAMAT101 FAT Papers), Module 5 accounts for **20 to 25 marks**, typically structured as two $5 + 5$ mark split questions or one $5 + 5$ question plus a 10-mark question on Separation of Variables or Lagrange's Multiplier Method.

Table 1: First-Order Nonlinear & Linear PDE Recognition & Solution Guide

PDE Type	Equation Form	Complete Integral Recipe
Type I (Only p, q)	$F(p, q) = 0$	Set $p = a, q = b \implies z = ax + by + c$, where $F(a, b) = 0 \implies b = \phi(a)$.
Type II (z, p, q only)	$F(z, p, q) = 0$	Set $u = x + ay$, replace $p = \frac{dz}{du}$ and $q = a\frac{dz}{du}$, then integrate separable ODE in z and u.
Type III (Separable)	$f_1(x, p) = f_2(y, q)$	Equate both sides to constant a: solve for $p = g_1(x, a)$ and $q = g_2(y, a)$, then $dz = \int p\,dx + \int q\,dy + c$.
Type IV (Clairaut's)	$z = px + qy + f(p, q)$	Complete Integral: Simply replace $p \to a, q \to b$: $z = ax + by + f(a, b)$. Singular Integral: Eliminate a, b using $x + \frac{\partial f}{\partial a} = 0$ and $y + \frac{\partial f}{\partial b} = 0$.
Lagrange's Linear	$Pp + Qq = R$	Solve auxiliary system $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$ to find two independent invariants $u(x, y, z) = c_1$ and $v(x, y, z) = c_2$; solution is $\phi(u, v) = 0$.

2 Topic 1: Formation of Partial Differential Equations

2.1 Theory: Eliminating Arbitrary Constants vs. Arbitrary Functions

1. **Eliminating Arbitrary Constants**: Given a relation $F(x, y, z, a, b) = 0$ with two arbitrary constants a, b, differentiate partially with respect to x (treating y as constant, noting $\frac{\partial z}{\partial x} = p$) and with respect to y (treating x as constant, noting $\frac{\partial z}{\partial y} = q$). Eliminate a and b among the three equations.
2. **Eliminating One Arbitrary Function** $\phi(u, v) = 0$: If $\phi\big(u(x, y, z),\, v(x, y, z)\big) = 0$, differentiating with respect to x and y and eliminating $\frac{\partial \phi}{\partial u}, \frac{\partial \phi}{\partial v}$ yields the **Jacobian Determinant Formula**:

$$\boxed{\begin{vmatrix} \dfrac{\partial u}{\partial x} + p\dfrac{\partial u}{\partial z} & \dfrac{\partial v}{\partial x} + p\dfrac{\partial v}{\partial z} \\ \dfrac{\partial u}{\partial y} + q\dfrac{\partial u}{\partial z} & \dfrac{\partial v}{\partial y} + q\dfrac{\partial v}{\partial z} \end{vmatrix} = 0.}$$

2.2 Worked VIT Question Bank & Past FAT Problems (Topic 1)

Problem 5.1.1 (VIT Question Bank Q14, Q15, Q24 & Tutorial Q1): *Form the PDE by eliminating the arbitrary constants from: (a)* $z = ax^2 + by^2$ *(b)* $(x - h)^2 + (y - k)^2 - z^2 = \lambda^2$ *(h, k arbitrary) (c)* $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

Solution:

Part (a): Differentiating $z = ax^2 + by^2$ with respect to x and y:

$$p = 2ax \implies a = \frac{p}{2x}, \qquad q = 2by \implies b = \frac{q}{2y}.$$

Substituting into $z = ax^2 + by^2$ gives $z = \left(\frac{p}{2x}\right)x^2 + \left(\frac{q}{2y}\right)y^2 \implies \boxed{xp + yq = 2z}$.

Part (b): Differentiating $(x-h)^2 + (y-k)^2 - z^2 = \lambda^2$ partially with respect to x and y:

$$2(x-h) - 2zp = 0 \implies x - h = zp, \qquad 2(y-k) - 2zq = 0 \implies y - k = zq.$$

Substituting back into the original equation gives $(zp)^2 + (zq)^2 - z^2 = \lambda^2 \implies \boxed{z^2(p^2 + q^2 - 1) = \lambda^2}$ (or $z^2(p^2 + q^2 + 1) = \lambda^2$ for sphere $(x-h)^2 + (y-k)^2 + z^2 = \lambda^2$).

Part (c): Differentiating $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ with respect to x and y:

$$\frac{x}{a^2} + \frac{zp}{c^2} = 0 \implies -\frac{c^2}{a^2} = \frac{zp}{x}, \qquad \frac{y}{b^2} + \frac{zq}{c^2} = 0 \implies -\frac{c^2}{b^2} = \frac{zq}{y}.$$

Differentiating the first relation wrt x gives $\frac{1}{a^2} + \frac{1}{c^2}(p^2 + zr) = 0 \implies xzr + xp^2 - zp = 0$ (or if $c = 1$ is fixed and only a, b are arbitrary, substituting $\frac{1}{a^2} = -\frac{zp}{x}, \frac{1}{b^2} = -\frac{zq}{y}$ into $\frac{x^2}{a^2} + \frac{y^2}{b^2} + z^2 = 1$ gives $\boxed{xzp + yzq = z^2 - 1}$). □

Problem 5.1.2 (BMAT102L FAT April 2025 Q3(ii) & VIT Question Bank Q20, Q7, Q23): *Form the PDE by eliminating the arbitrary function(s): (a) $f(ax + by + cz,\ x^2 + y^2 + z^2) = 0$ (b) $\phi(x^2 + y^2 + z^2,\ z^2 - 2xy) = 0$ (c) $z = f(x^2 - y) + g(x^2 + y)$.*

Solution:

Part (a) [FAT April 2025]: Let $u = ax + by + cz$ and $v = x^2 + y^2 + z^2$. Using the Jacobian formula:

$$\begin{vmatrix} a + cp & 2x + 2zp \\ b + cq & 2y + 2zq \end{vmatrix} = 0 \implies \frac{a + cp}{b + cq} = \frac{x + zp}{y + zq} \implies \boxed{(cy - bz)p + (az - cx)q = bx - ay.}$$

Part (b) [VIT QB Q20]: Let $u = x^2 + y^2 + z^2$ and $v = z^2 - 2xy$:

$$\begin{vmatrix} 2x + 2zp & -2y + 2zp \\ 2y + 2zq & -2x + 2zq \end{vmatrix} = 0 \implies \boxed{(x + zp)(zq - x) = (y + zq)(zp - y).}$$

Part (c) [VIT QB Q7]: Differentiating $z = f(x^2 - y) + g(x^2 + y)$:

$$z_x = 2xf'(x^2 - y) + 2xg'(x^2 + y) \implies \frac{z_x}{2x} = f'(x^2 - y) + g'(x^2 + y),$$

$$z_{xx} = 2(f' + g') + 4x^2(f''(x^2 - y) + g''(x^2 + y)) = \frac{1}{x}z_x + 4x^2 z_{yy} \quad \text{since } z_{yy} = f'' + g''.$$

Hence $\boxed{z_{xx} = \frac{1}{x}z_x + 4x^2 z_{yy}}$. □

3 Topic 2: Standard Types of First-Order Nonlinear PDEs – 1 (Types I & II)

3.1 Theory: Type I $F(p, q) = 0$ and Type II $F(z, p, q) = 0$

1. **Standard Type I ($F(p, q) = 0$):** Since x, y, z do not appear explicitly, the **Complete Integral** is:

$$\boxed{z = ax + by + c, \qquad \text{where } F(a, b) = 0.}$$

Solving $F(a, b) = 0$ for $b = \phi(a)$ gives $z = ax + \phi(a)y + c$ (a 2-parameter family with parameters a, c).

2. **Standard Type II** ($F(z,p,q)=0$): Assume a traveling-wave trial solution $z=f(u)$ where $\boxed{u=x+ay}$. Then $p=\frac{\partial z}{\partial x}=\frac{dz}{du}$ and $q=\frac{\partial z}{\partial y}=a\frac{dz}{du}$. Substituting into $F\left(z,\frac{dz}{du},a\frac{dz}{du}\right)=0$ yields a first-order ODE for $\frac{dz}{du}=g(z,a)$, integrated as $\int\frac{dz}{g(z,a)}=x+ay+b$.

3.2 Worked VIT Question Bank & Past FAT Problems (Topic 2)

Problem 5.2.1 (VIT Question Bank Q4 & Q19): *Find the complete integral of: (a)* $p^2+q^2=npq$ *and (b)* $2p-3q=pq$.

Solution:

Part (a): This is Type I ($F(p,q)=0$). Set $p=a,q=b$ in $z=ax+by+c$:

$$a^2+b^2-nab=0 \implies b=\frac{na\pm\sqrt{n^2a^2-4a^2}}{2}=\frac{a}{2}\left(n\pm\sqrt{n^2-4}\right).$$

Thus the complete integral is $\boxed{z=ax+\frac{a}{2}\left(n\pm\sqrt{n^2-4}\right)y+c}$.

Part (b): Setting $p=a,q=b$ gives $2a-3b=ab \implies b=\frac{2a}{a+3}$, so $\boxed{z=ax+\frac{2a}{a+3}y+c}$. □

Problem 5.2.2 (BMAT102L FAT April 2025 Q4(ii) & VIT Question Bank Q16, Q22): *Find the complete and singular integrals of: (a)* $9(p^2z+q^2)=4$ *(b)* $p^2+q^2=z^2$ *(c)* $q^2=z^2p^2(1-p^2)$.

Solution:

All three equations contain only z,p,q (**Type II**), so let $u=x+ay$, $p=\frac{dz}{du}$, $q=a\frac{dz}{du}$.

Part (a) [FAT April 2025]: Substituting $p=\frac{dz}{du},q=a\frac{dz}{du}$ into $9(zp^2+q^2)=4$:

$$9(z+a^2)\left(\frac{dz}{du}\right)^2=4 \implies \int 3\sqrt{z+a^2}\,dz=\pm2u+b \implies \boxed{2(z+a^2)^{3/2}=\pm2(x+ay)+b.}$$

Part (b) [VIT QB Q16]: $(1+a^2)\left(\frac{dz}{du}\right)^2=z^2 \implies \frac{dz}{z}=\pm\frac{du}{\sqrt{1+a^2}} \implies \ln z=\pm\frac{x+ay}{\sqrt{1+a^2}}+\ln C$, or equivalently $\boxed{z=Ce^{\alpha x+\beta y}}$ where $\alpha^2+\beta^2=1$. Differentiating wrt C gives $e^{\alpha x+\beta y}=0$, so **no singular integral exists** (or $z=0$ trivial).

Part (c) [VIT QB Q22]: $a^2\left(\frac{dz}{du}\right)^2=z^2\left(\frac{dz}{du}\right)^2\left[1-\left(\frac{dz}{du}\right)^2\right] \implies \left(\frac{dz}{du}\right)^2=\frac{z^2-a^2}{z^2} \implies \frac{z\,dz}{\sqrt{z^2-a^2}}=\pm du.$

Integrating gives $\sqrt{z^2-a^2}=\pm(x+ay+b) \implies \boxed{(x+ay+b)^2=z^2-a^2}$ (Complete Integral), and $z=0$ is the Singular Integral. □

4 Topic 3: Standard Types of First-Order Nonlinear PDEs – 2 (Types III, IV & Reducible)

4.1 Theory: Type III (Separable), Type IV (Clairaut's Form) & Reducible Equations

1. **Standard Type III** ($f_1(x,p)=f_2(y,q)$): Equate both sides to an arbitrary constant a:

$$f_1(x,p)=f_2(y,q)=a \implies p=g_1(x,a),\quad q=g_2(y,a) \implies \boxed{z=\int g_1(x,a)\,dx+\int g_2(y,a)\,dy+c.}$$

2. **Standard Type IV (Clairaut's PDE** $z=px+qy+f(p,q)$**)**:
   - **Complete Integral**: Simply replace $p\to a$ and $q\to b$: $\boxed{z=ax+by+f(a,b)}$.

- **Singular Integral (Envelope)**: Differentiate the Complete Integral partially with respect to a and b:

$$\frac{\partial z}{\partial a} = x + \frac{\partial f}{\partial a} = 0 \implies \frac{\partial f}{\partial a} = -x, \qquad \frac{\partial z}{\partial b} = y + \frac{\partial f}{\partial b} = 0 \implies \frac{\partial f}{\partial b} = -y,$$

and eliminate a and b to obtain $z(x, y)$ containing no arbitrary constants!

4.2 Worked VIT Question Bank & Past FAT Problems (Topic 3)

Problem 5.3.1 (BMAT102L FAT April 2025 Q3(i) & VIT Question Bank Q19): *Find the complete and singular solutions of the Clairaut PDEs: (a)* $px - z = 2\sqrt{pq} - yq$ *and (b)* $z = px + qy + p^2q^2$.

Solution:

Part (a) [FAT April 2025]: Rearranging $px - z = 2\sqrt{pq} - yq$ gives Clairaut's form:

$$z = px + qy - 2\sqrt{pq}.$$

1. **Complete Integral**: Replace $p \to a, q \to b$:

$$\boxed{z = ax + by - 2\sqrt{ab}.}$$

2. **Singular Integral**: Differentiate with respect to a and b:

$$\frac{\partial z}{\partial a} = x - \sqrt{\frac{b}{a}} = 0 \implies x = \sqrt{\frac{b}{a}}, \qquad \frac{\partial z}{\partial b} = y - \sqrt{\frac{a}{b}} = 0 \implies y = \sqrt{\frac{a}{b}}.$$

Multiplying these two equations eliminates a and b immediately: $\boxed{xy = 1}$ (and substituting into z gives $z = 0$ on $xy = 1$).

Part (b) [VIT QB Q19]: For $z = px + qy + p^2q^2$: 1. **Complete Integral**: $\boxed{z = ax + by + a^2b^2}$. 2. **Singular Integral**: Differentiate wrt a and b:

$$x + 2ab^2 = 0 \implies ab^2 = -\frac{x}{2}, \qquad y + 2a^2b = 0 \implies a^2b = -\frac{y}{2}.$$

Multiplying gives $a^3b^3 = \frac{xy}{4} \implies ab = (\frac{xy}{4})^{1/3}$. Since $ax = -2a^2b^2$ and $by = -2a^2b^2$, we have:

$$z = (-2a^2b^2) + (-2a^2b^2) + a^2b^2 = -3a^2b^2 = -3\left(\frac{xy}{4}\right)^{2/3} = \boxed{-\frac{3}{4}(4)^{1/3}x^{2/3}y^{2/3}.} \quad \square$$

Problem 5.3.2 (VIT Tutorial Sheet Q3 & Q5): *(a) Solve* $p\sqrt{x} + q\sqrt{y} = \sqrt{z}$. *(b) Solve* $xp - yq = y^2 - x^2$.

Solution:

Part (a): Can be solved either as Type III after dividing by $\sqrt{z}$ or by Lagrange's method: Separating variables (or transforming $Z = 2\sqrt{z}, X = 2\sqrt{x}, Y = 2\sqrt{y}$): by Lagrange's auxiliary equations $\frac{dx}{\sqrt{x}} = \frac{dy}{\sqrt{y}} = \frac{dz}{\sqrt{z}}$, integrating pairs gives $\sqrt{x} - \sqrt{y} = c_1$ and $\sqrt{y} - \sqrt{z} = c_2$, so $\boxed{\phi\left(\sqrt{x} - \sqrt{y},\ \sqrt{y} - \sqrt{z}\right) = 0}$.

Part (b): Rearranging $xp - yq = y^2 - x^2$ into Type III separable form:

$$xp + x^2 = yq + y^2 = k \implies p = \frac{k - x^2}{x}, \qquad q = \frac{k - y^2}{y}.$$

Integrating $dz = p\,dx + q\,dy$:

$$z = \int \left(\frac{k}{x} - x\right) dx + \int \left(\frac{k}{y} - y\right) dy + c \implies \boxed{z = k\ln(xy) - \frac{1}{2}(x^2 + y^2) + c.} \quad \square$$

5 Topic 4 & Topic 5: Lagrange's Linear Equation ($Pp + Qq = R$)

5.1 Theory: Method of Grouping (Topic 4) & Method of Multipliers (Topic 5)

A first-order quasi-linear PDE has the form $\boxed{P(x,y,z)p + Q(x,y,z)q = R(x,y,z)}$. Its **Lagrange Auxiliary (Subsidiary) Equations** are:

$$\boxed{\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}.}$$

To find two independent integrals $u(x,y,z) = c_1$ and $v(x,y,z) = c_2$ (so that the general solution is $\phi(u,v) = 0$):

1. **Method 1: Method of Grouping (Topic 4)**: Pair any two ratios directly (e.g., $\frac{dx}{P} = \frac{dy}{Q}$) if the third variable (z) cancels out or can be substituted using an already-found first integral $u(x,y,z) = c_1$.
2. **Method 2: Method of Multipliers (Topic 5)**: By algebra of equal ratios, for any multipliers (l,m,n):
$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l\,dx + m\,dy + n\,dz}{lP + mQ + nR}.$$
Golden Rule: If you choose (l,m,n) such that the denominator $\boxed{lP + mQ + nR = 0}$, then the numerator must also be zero: $\boxed{l\,dx + m\,dy + n\,dz = 0}$, which integrates directly to give $u(x,y,z) = c_1$! *Standard VIT Multiplier Sets to Test First*: $(1,1,1)$, (x,y,z), $\left(\frac{1}{x}, \frac{1}{y}, \frac{1}{z}\right)$, and (yz, zx, xy).

5.2 Worked VIT Question Bank & Past FAT Problems (Topics 4 & 5)

Problem 5.4.1 (BMAT102L FAT April 2025 Q4(i) & VIT Question Bank Q1(iv)): *Solve the Lagrange linear PDEs: (a)* $x^2p - y^2q = z(x-y)$ *and (b)* $y^2p - xyq = x(z-2y)$.

Solution:

Part (a) [FAT April 2025]: Auxiliary equations are $\frac{dx}{x^2} = \frac{dy}{-y^2} = \frac{dz}{z(x-y)}$.

- **First Integral (Grouping first two)**: $\int \frac{dx}{x^2} = -\int \frac{dy}{y^2} \implies -\frac{1}{x} = \frac{1}{y} - c_1 \implies \frac{1}{x} + \frac{1}{y} = c_1$.
- **Second Integral (Multipliers $l = 1, m = 1, n = 0$)**: $\frac{dx+dy}{x^2-y^2} = \frac{dz}{z(x-y)} \implies \frac{dx+dy}{x+y} = \frac{dz}{z} \implies \ln|x+y| = \ln|z| + \ln c_2 \implies \frac{x+y}{z} = c_2$.

General solution: $\boxed{\phi\left(\frac{1}{x} + \frac{1}{y}, \frac{x+y}{z}\right) = 0}$.

Part (b) [VIT QB Q1(iv)]: Auxiliary equations: $\frac{dx}{y^2} = \frac{dy}{-xy} = \frac{dz}{x(z-2y)}$.

- **First Integral (First two ratios)**: $\frac{dx}{y} = \frac{dy}{-x} \implies x\,dx + y\,dy = 0 \implies x^2 + y^2 = c_1$.
- **Second Integral (Second and third ratios)**: $\frac{dy}{-y} = \frac{dz}{z-2y} \implies y\,dz + z\,dy - 2y\,dy = 0 \implies d(yz) - d(y^2) = 0 \implies yz - y^2 = c_2$.

General solution: $\boxed{\phi(x^2 + y^2,\ yz - y^2) = 0}$. □

Problem 5.5.1 (VIT Question Bank Q1(i), (iii), (v), (vi) — Multiplier Masterclass): *Solve using Lagrange's Method of Multipliers: (a)* $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$ *(b)* $x(y^2 - z^2)p + y(z^2 - x^2)q = z(y^2 - x^2)$ *(c)* $\left(\frac{y-z}{yz}\right)p + \left(\frac{z-x}{zx}\right)q = \frac{x-y}{xy}$.

Solution:

Part (a): Auxiliary equations: $\frac{dx}{x^2-yz} = \frac{dy}{y^2-zx} = \frac{dz}{z^2-xy}$. Taking differences of pairs gives:

$$\frac{dx-dy}{(x-y)(x+y+z)} = \frac{dy-dz}{(y-z)(x+y+z)} \implies \frac{d(x-y)}{x-y} = \frac{d(y-z)}{y-z} \implies \frac{x-y}{y-z} = c_1.$$

Using multipliers (x, y, z) and $(1, 1, 1)$ gives $\frac{x\,dx+y\,dy+z\,dz}{x^3+y^3+z^3-3xyz} = \frac{dx+dy+dz}{x^2+y^2+z^2-xy-yz-zx}$, which simplifies to $d(xy+yz+zx) = 0 \implies xy+yz+zx = c_2$. General solution: $\boxed{\phi\left(xy+yz+zx,\ \frac{x-y}{y-z}\right) = 0}$.

Part (b): Auxiliary equations: $\frac{dx}{x(y^2-z^2)} = \frac{dy}{y(z^2-x^2)} = \frac{dz}{z(y^2-x^2)}$ (note: for cyclic $z(x^2-y^2)$):

- **Multipliers** (x, y, z): Denominator $x^2(y^2-z^2)+y^2(z^2-x^2)+z^2(x^2-y^2) = 0 \implies x\,dx+y\,dy+z\,dz = 0 \implies x^2+y^2+z^2 = c_1$.
- **Multipliers** $(1/x, 1/y, 1/z)$: Denominator $(y^2-z^2)+(z^2-x^2)+(x^2-y^2) = 0 \implies \frac{dx}{x}+\frac{dy}{y}+\frac{dz}{z} = 0 \implies xyz = c_2$.

General solution: $\boxed{\phi(x^2+y^2+z^2,\ xyz) = 0}$.

Part (c): Multiply through by xyz: $x(y-z)p + y(z-x)q = z(x-y)$.

- **Multipliers** $(1, 1, 1)$: $x(y-z)+y(z-x)+z(x-y) = 0 \implies dx+dy+dz = 0 \implies x+y+z = c_1$.
- **Multipliers** $(1/x, 1/y, 1/z)$: $(y-z)+(z-x)+(x-y) = 0 \implies \frac{dx}{x}+\frac{dy}{y}+\frac{dz}{z} = 0 \implies xyz = c_2$.

General solution: $\boxed{\phi(x+y+z,\ xyz) = 0}$. □

6 Topic 6: Solution of a Linear First-Order PDE by Separation of Variables

6.1 Theory & Superposition Step-by-Step Recipe

To solve a linear PDE in $u(x, y)$ such as $a\frac{\partial u}{\partial x} + b\frac{\partial u}{\partial y} + cu = 0$ subject to a boundary condition like $u(0, y) = Ae^{m_1 y} + Be^{m_2 y}$:

1. **Assume Product Form**: Set $\boxed{u(x, y) = X(x)Y(y)}$, so $\frac{\partial u}{\partial x} = X'(x)Y(y)$ and $\frac{\partial u}{\partial y} = X(x)Y'(y)$.
2. **Divide by $X(x)Y(y)$ and Separate**: Group all functions of x on the LHS and all functions of y on the RHS, and set both sides equal to a separation constant k.
3. **Solve the Two ODEs**: Obtain $X(x) = C_1 e^{\alpha(k)x}$ and $Y(y) = C_2 e^{\beta(k)y}$, so $u_k(x, y) = Ce^{\alpha(k)x+\beta(k)y}$.
4. **Apply Boundary Condition via Superposition**: If the boundary condition has two exponential terms (e.g., $u(0, y) = 3e^{-y} - e^{-5y}$), write a two-term sum $u(x, y) = C_1 e^{\alpha(k_1)x+\beta(k_1)y} + C_2 e^{\alpha(k_2)x+\beta(k_2)y}$ and match each exponent separately!

6.2 Worked VIT Question Bank Problems (Topic 6)

Problem 5.6.1 (VIT Question Bank Q2 & Q21 — High-Frequency 10-Mark Problem): *Using the method of separation of variables, solve* $4\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u$, *given that: (a)* $u(0, y) = e^{-5y}$ *and (b)* $u(0, y) = 3e^{-y} - e^{-5y}$.

Solution:

1. **Substitute** $u(x,y) = X(x)Y(y)$:

$$4X'(x)Y(y) + X(x)Y'(y) = 3X(x)Y(y).$$

Divide throughout by $X(x)Y(y)$ and separate variables:

$$4\frac{X'}{X} + \frac{Y'}{Y} = 3 \implies \frac{4X' - 3X}{X} = -\frac{Y'}{Y} = k \quad (\text{or } \frac{Y'}{Y} = k).$$

Let us set $\frac{Y'}{Y} = k \implies Y(y) = C_2 e^{ky}$. Then:

$$4\frac{X'}{X} = 3 - k \implies \frac{X'}{X} = \frac{3-k}{4} \implies X(x) = C_1 e^{\left(\frac{3-k}{4}\right)x}.$$

Thus the separated building-block solution is:

$$\boxed{u_k(x,y) = Ce^{\left(\frac{3-k}{4}\right)x+ky}.}$$

2. **Part (a): Match** $u(0,y) = e^{-5y}$: Setting $x = 0$ gives $u(0,y) = Ce^{ky} = 1 \cdot e^{-5y} \implies C = 1$ and $k = -5$. Then $\frac{3-k}{4} = \frac{3-(-5)}{4} = 2$. Hence:

$$\boxed{u(x,y) = e^{2x-5y}.}$$

3. **Part (b): Match** $u(0,y) = 3e^{-y} - e^{-5y}$ **by Superposition**: Write $u(x,y) = C_1 e^{\left(\frac{3-k_1}{4}\right)x+k_1 y} + C_2 e^{\left(\frac{3-k_2}{4}\right)x+k_2 y}$. At $x = 0$:

- First term: $C_1 = 3,\ k_1 = -1 \implies \frac{3-(-1)}{4} = 1 \implies 3e^{x-y}$.
- Second term: $C_2 = -1,\ k_2 = -5 \implies \frac{3-(-5)}{4} = 2 \implies -e^{2x-5y}$.

Combining both modes gives:

$$\boxed{u(x,y) = 3e^{x-y} - e^{2x-5y}.} \quad \square$$

Problem 5.6.2 (VIT Question Bank Q9, Q10 & Q11): *Solve by separation of variables: (a)* $\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} - 2u$, *given* $u(x,0) = 10e^{-x} - 6e^{-4x}$. *(b)* $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} + u$, *given* $u(x,0) = 3e^{-5x} + 2e^{-3x}$. *(c)* $3\frac{\partial u}{\partial x} + 2\frac{\partial u}{\partial y} = 0$, *given* $u(x,0) = 6e^{-2x}$.

Solution:

Part (a): Let $u(x,t) = X(x)T(t) \implies \frac{T'}{T} = \frac{X'}{X} - 2$. Set $\frac{X'}{X} = k \implies X = e^{kx}$ and $\frac{T'}{T} = k - 2 \implies T = e^{(k-2)t}$, so $u_k(x,t) = Ce^{kx+(k-2)t}$. Matching $u(x,0) = 10e^{-x} - 6e^{-4x}$:

- For $k_1 = -1, C_1 = 10$: $k_1 - 2 = -3 \implies 10e^{-x-3t}$.
- For $k_2 = -4, C_2 = -6$: $k_2 - 2 = -6 \implies -6e^{-4x-6t}$.

Thus $\boxed{u(x,t) = 10e^{-x-3t} - 6e^{-4x-6t}}$.

Part (b): $u_k(x,y) = Ce^{kx+(k-1)y}$. With $k_1 = -5$ $(k_1 - 1 = -6)$ and $k_2 = -3$ $(k_2 - 1 = -4)$, $\boxed{u(x,y) = 3e^{-5x-6y} + 2e^{-3x-4y}}$ (or if $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} - u$, $\boxed{3e^{-5x-4y} + 2e^{-3x-2y}}$).

Part (c): $3\frac{X'}{X} = -2\frac{Y'}{Y}$. For $X(x) = e^{-2x}$ $(\frac{X'}{X} = -2)$, we get $-2\frac{Y'}{Y} = 3(-2) = -6 \implies \frac{Y'}{Y} = 3 \implies Y(y) = e^{3y}$. Thus $\boxed{u(x,y) = 6e^{3y-2x}}$. $\square$

7 Topic 7 & Topic 8: Lagrange's Equation for Flow Modelling & Separation of Variables for CPU Temperature

7.1 Topic 7: Lagrange's Equation for Flow Modelling (Advection & Transport)

In fluid dynamics, pollutant dispersion, and traffic flow modelling, the concentration or density $c(x,t)$ of a quantity carried by a velocity field $v(x,t)$ with source/sink decay rate $S(x,t,c)$ satisfies the **first-order advection-reaction PDE** (a direct physical application of Lagrange's linear equation!):

$$\boxed{\frac{\partial c}{\partial t} + v(x,t)\frac{\partial c}{\partial x} = S(x,t,c) \iff 1 \cdot \frac{\partial c}{\partial t} + v(x,t)\frac{\partial c}{\partial x} = S.}$$

Its Lagrange auxiliary equations (**Method of Characteristics**) are:

$$\frac{dt}{1} = \frac{dx}{v(x,t)} = \frac{dc}{S(x,t,c)}.$$

The first equation $\frac{dx}{dt} = v(x,t)$ defines the **characteristic fluid particle trajectories** $x - vt = c_1$, while $\frac{dc}{dt} = S$ describes how the concentration changes along each moving fluid parcel!

Problem 5.7.1 (Applied Flow Modelling Problem): *A chemical tracer in a 1D microfluidic channel with constant flow velocity v_0 and first-order decay rate λ obeys the advection-decay PDE $\frac{\partial c}{\partial t} + v_0\frac{\partial c}{\partial x} = -\lambda c$, subject to the initial concentration profile $c(x,0) = C_0 e^{-x^2}$. Solve for $c(x,t)$ using Lagrange's auxiliary equations.*

Solution:

1. **Lagrange Auxiliary Equations**:
$$\frac{dt}{1} = \frac{dx}{v_0} = \frac{dc}{-\lambda c}.$$
2. **Find the Two Characteristic Invariants**:

- From $\frac{dt}{1} = \frac{dx}{v_0}$: $dx - v_0\,dt = 0 \implies x - v_0 t = k_1$.
- From $\frac{dt}{1} = \frac{dc}{-\lambda c}$: $\frac{dc}{c} = -\lambda\,dt \implies ce^{\lambda t} = k_2$.

Thus the general solution is $c(x,t)e^{\lambda t} = f(x - v_0 t) \implies c(x,t) = e^{-\lambda t} f(x - v_0 t)$. 3. **Apply Initial Profile** $c(x,0) = C_0 e^{-x^2}$: At $t = 0$, $f(x) = C_0 e^{-x^2}$, so replacing $x \to x - v_0 t$ yields the damped traveling wave:

$$\boxed{c(x,t) = C_0 e^{-\lambda t} e^{-(x - v_0 t)^2}.} \quad \square$$

7.2 Topic 8: Separation of Variables for Analyzing CPU Temperature

In microprocessor and heat-sink thermal engineering, the temperature excess $u(x,t) = T(x,t) - T_{\text{ambient}}$ along a cooling fin or silicon die of thermal diffusivity $\alpha = \frac{k}{\rho c_p}$ with convective surface heat loss γ satisfies the 1D reaction-diffusion heat PDE (featured in **VIT Tutorial Sheet Q8, Q9, Q10**):

$$\boxed{au_{xx} - bu_t + cu = 0 \qquad \text{or} \qquad \frac{\partial u}{\partial t} = \alpha\frac{\partial^2 u}{\partial x^2} - \gamma u.}$$

Problem 5.8.1 (VIT Tutorial Sheet Q8, Q9 & Q10 — CPU Fin Thermal PDE Reduction & Separation): *(a) Consider the thermal PDE $au_{xx} - bu_t + cu = 0$ $(b \neq 0)$. Let $u(x,t) = e^{\delta t} w(x,t)$. Find δ and K so that $w(x,t)$ satisfies the standard heat equation $w_t = Kw_{xx}$. (b) Solve the steady-decay thermal PDE $\frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial t} = 0$ by separation of variables so that temperature $u(x,t) \to 0$ as $t \to \infty$.*

Solution:

Part (a) [VIT Tutorial Sheet Q10]: Differentiating $u(x,t) = e^{\delta t} w(x,t)$:

$$u_{xx} = e^{\delta t} w_{xx}, \qquad u_t = e^{\delta t}\left(w_t + \delta w\right).$$

Substituting into $au_{xx} - bu_t + cu = 0$ and cancelling $e^{\delta t} \neq 0$:

$$aw_{xx} - b(w_t + \delta w) + cw = 0 \implies aw_{xx} - bw_t + (c - b\delta)w = 0.$$

To eliminate the w term, choose $\boxed{\delta = \dfrac{c}{b}}$, which reduces the PDE to $\boxed{w_t = K w_{xx}}$ with thermal diffusivity $\boxed{K = \dfrac{a}{b}}$.

Part (b) [VIT Tutorial Sheet Q8]: Substitute $u(x,t) = X(x)T(t)$ into $u_{xx} - u_t = 0$:

$$X''(x)T(t) = X(x)T'(t) \implies \frac{X''(x)}{X(x)} = \frac{T'(t)}{T(t)} = -p^2 \quad (\text{so } T(t) \to 0 \text{ as } t \to \infty).$$

Solving the two ODEs $X'' + p^2 X = 0$ and $T' + p^2 T = 0$ gives the physical cooling solution:

$$\boxed{u(x,t) = C_1 e^{-p^2 t}\left(C_2 \cos px + C_3 \sin px\right).} \quad \square$$